

Implementing Multi-Party Computation from Quantum Oblivious Transfers

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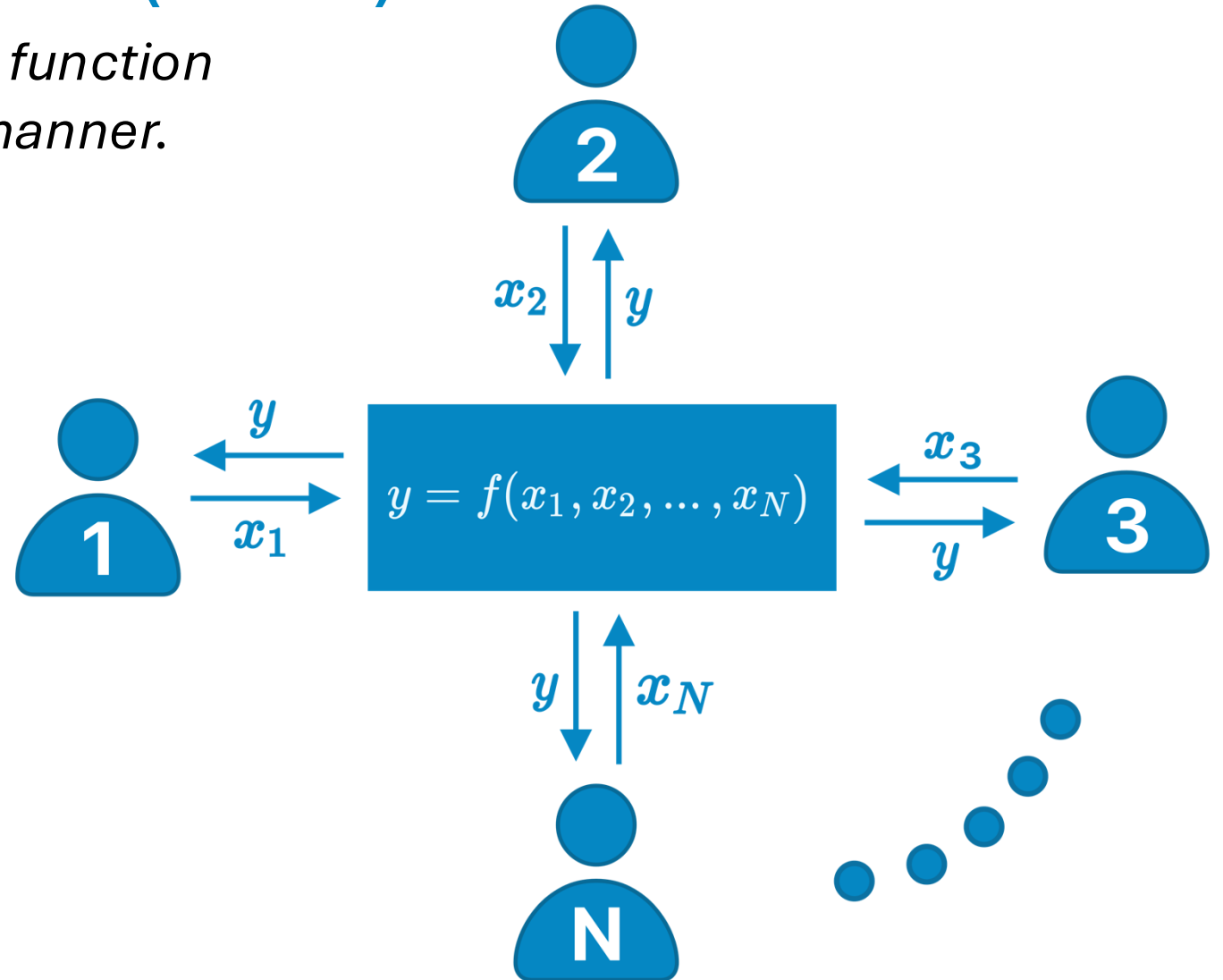
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SURF

Multiparty Computation (MPC)

Carrying out a joint computation of a function among multiple parties in a secure manner.

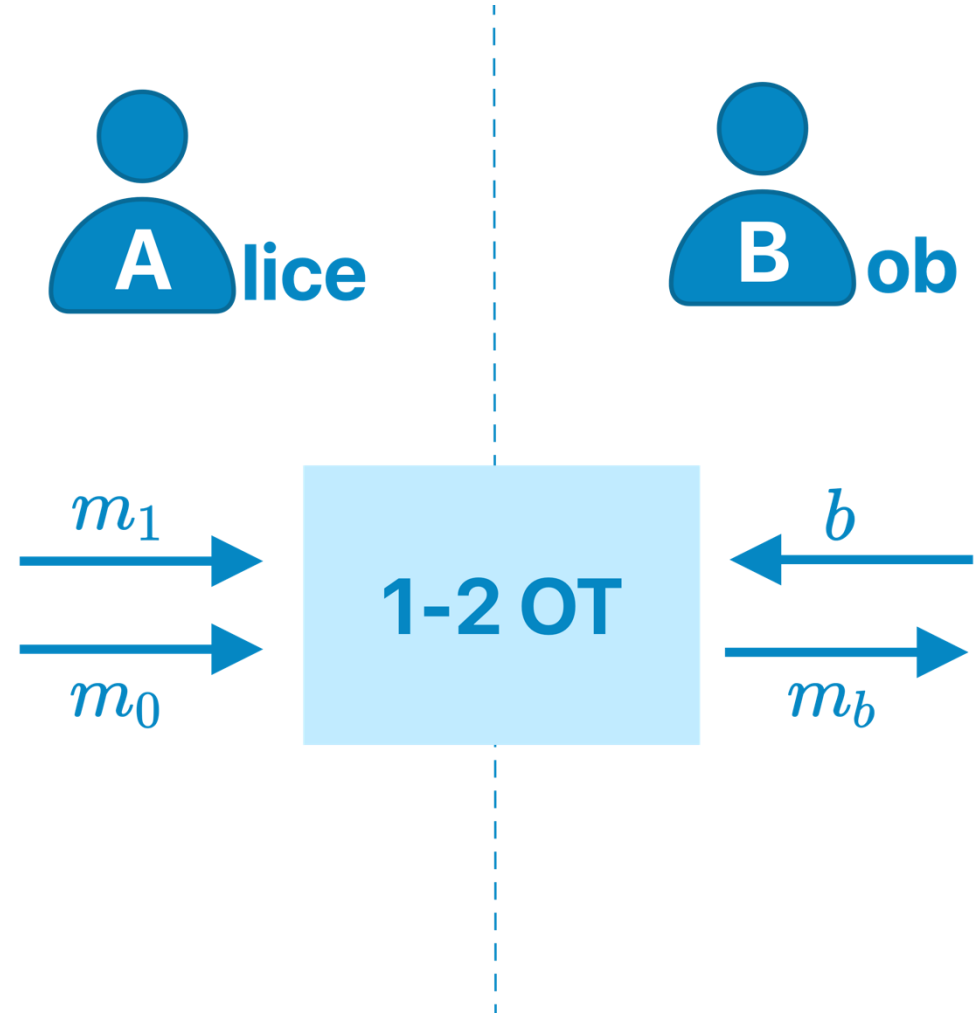
- No additional data known beyond what can be derived from the result



Oblivious Transfer (OT)

A sender transfers multiple pieces of information, but remains oblivious as to what has been transferred

- Key building block for MPC protocols
- Requires PKI
 - Relies on computational hardness of Asymmetric encryption
- Different variants:
 - Bit, 1-2, 1-N
 - Rabin



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Are you Shor about that?

- Several computational hard problems can be broken by Quantum computation.
- Breaking asymmetric encryption
-> PKI -> OT -> MPC

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Quantum Cryptography

Quantum mechanics makes it is impossible to measure an arbitrary quantum system's state without disturbing it, making eavesdropping detectable.

- Quantum Key Distribution
- Two communication techniques:
 - **Prepare-and-measure**
 - Transmission of individual quantum states from the sender to the receiver
 - **Entanglement-based**
 - Measurement outcomes are correlated, even when spatially separated

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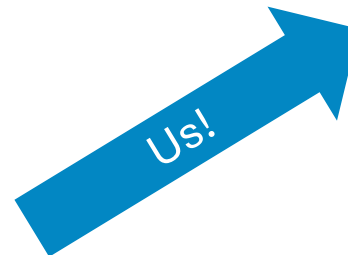
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Simulation Environment

Testing for **current** and **optimistic** hardware

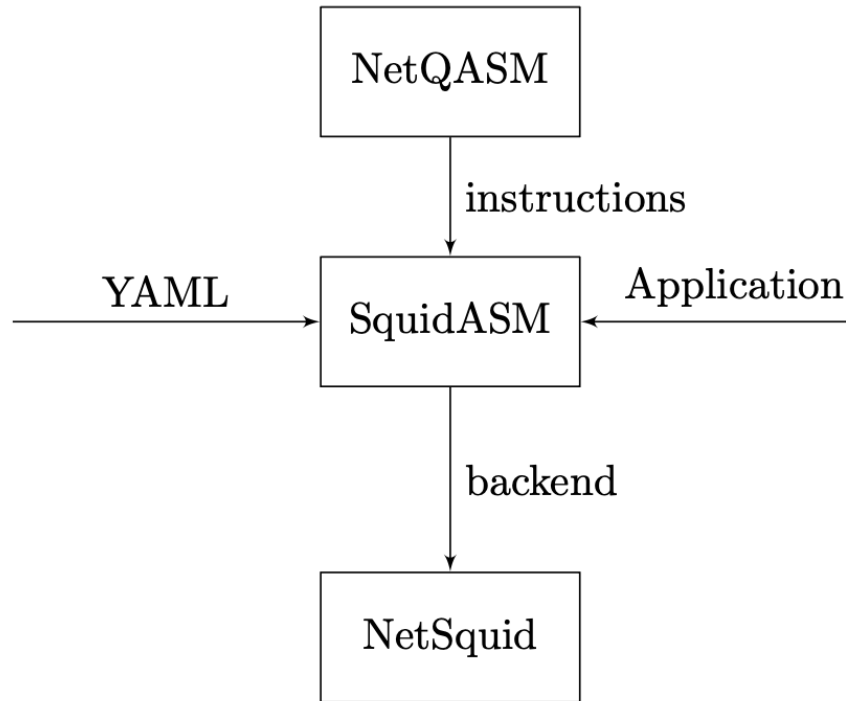


Figure 4.1: The used simulation stack.

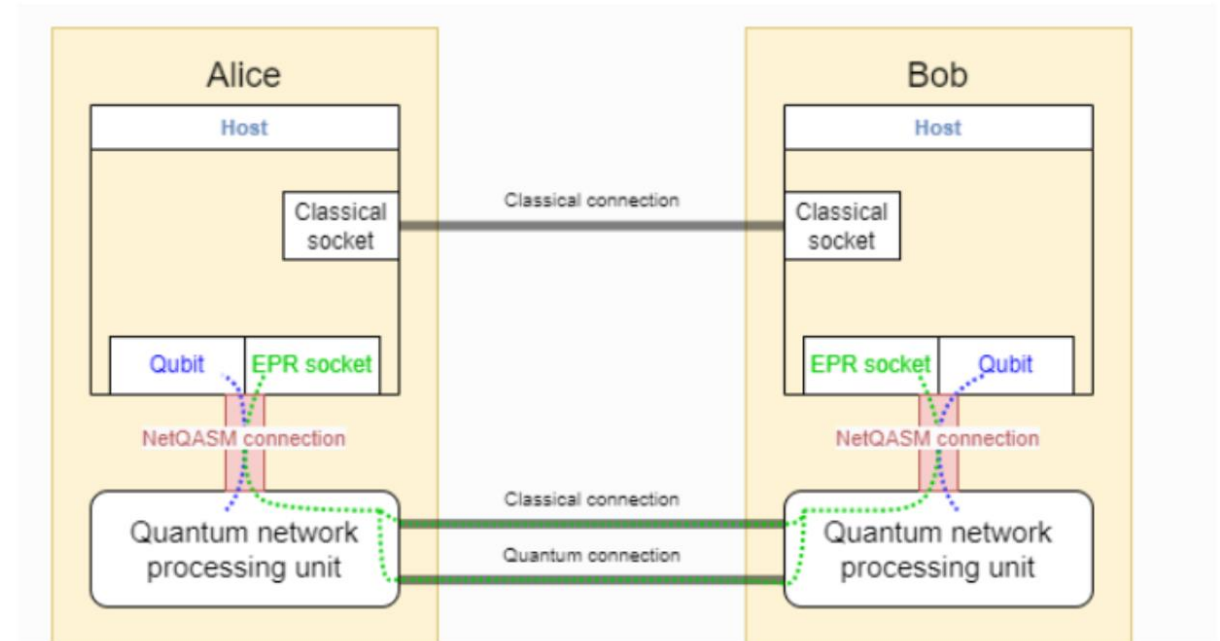


Figure 4.2: Schematic representation of the network setup using a NetQASM connection.

Quantum Oblivious Transfer (QOT)

Step 1: State Distribution

Alice

Input: Messages (m_0, m_1)

Common input: λ_{OT} , λ_{PQS} , $h_i()$, $PRG()$, $Comm()$

1.a Chooses $\theta_A \xleftarrow{\$} \{+, \times\}^{2\lambda_{OT}}$, and prepares EPR pair (2 qubits $|x_{AB}\rangle$), of which she measures $|x_A^{\theta_A}\rangle$ and sends bob x_B .

Bob

Input: Bit $b \in \{0, 1\}$

1.b Measures his qubit with random basis $\hat{\theta} \xleftarrow{\$} \{+, \times\}^{2\lambda_{OT}}$ to obtain $\hat{x} \in \{0, 1\}^{2\lambda_{OT}}$.

Entangled States



Quantum Oblivious Transfer (QOT)

Step 2: Bit commitment

Alice

2. Receives commitment t .

3. Chooses random subset $T \subset [2\lambda_{OT}]$ with $|T| = \lambda_{OT}$ and sends T .

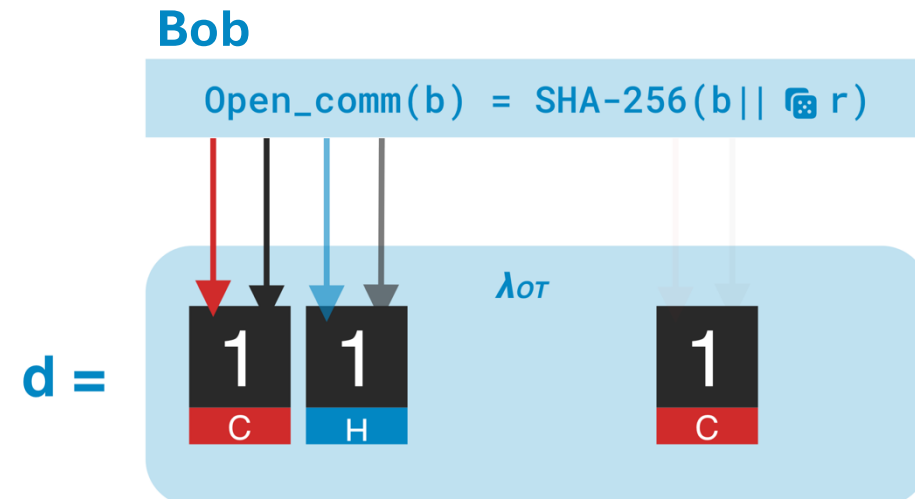
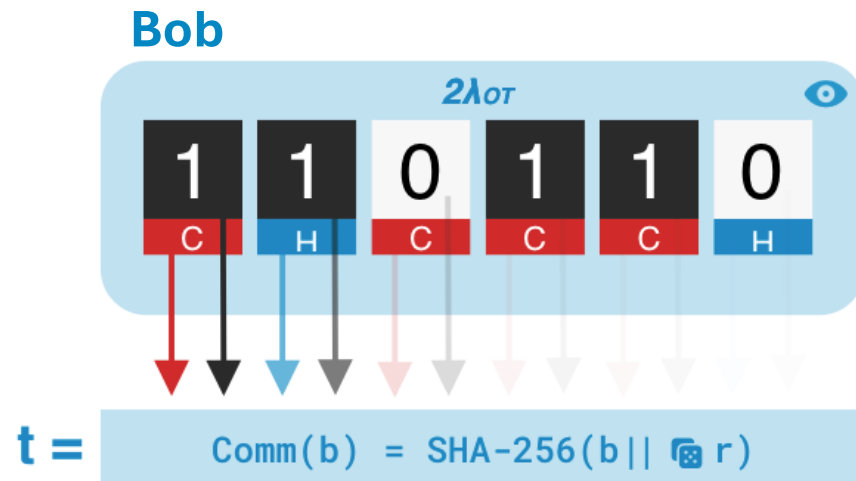
4. Checks $1 - \text{Verif}(t, d)/\lambda_{OT} \leq Q_{\text{tol}}$, otherwise aborts.

Bob

2. Sends public commitment $t = \text{Comm}(\hat{x}) + \text{Comm}(\hat{\theta})$.

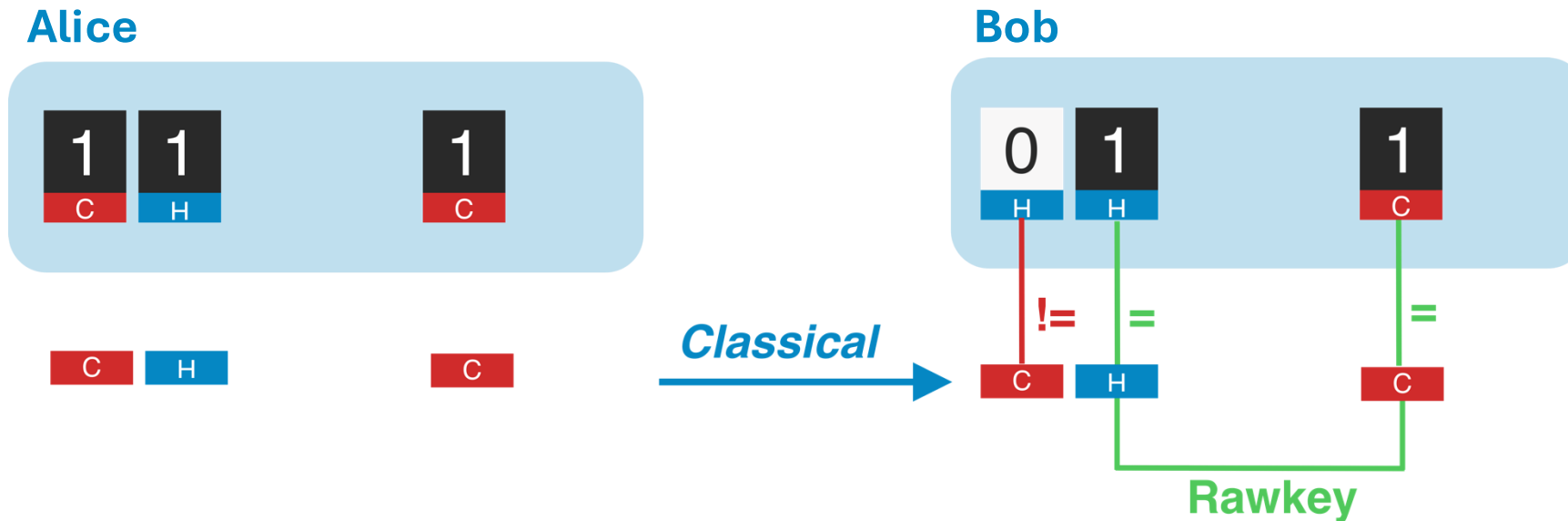
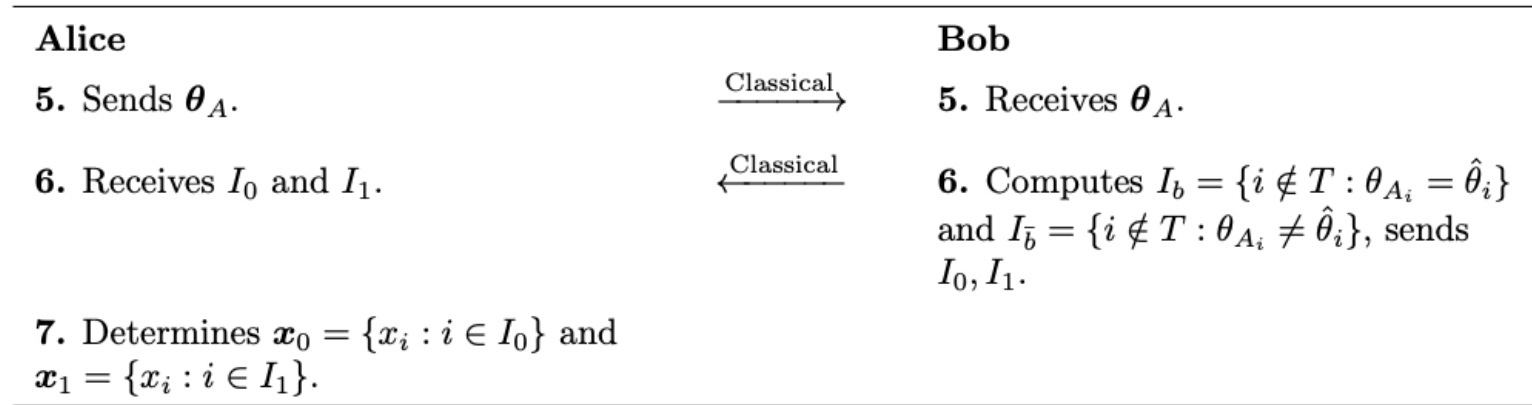
3. Receives T .

4. Opens and sends $d = \text{Open_comm}(\hat{x}_T) + \text{Open_Comm}(\hat{\theta}_T)$.



Quantum Oblivious Transfer (QOT)

Step 3: Basis Reconciliation



Quantum Oblivious Transfer (QOT)

Step 4: Error Correction

Alice	Bob
8. Calculates QBER using Bob's exposed bitstrings.	
Generates and sends $\sigma_0 = \text{Synd}(\mathbf{x}_0)$, $\sigma_1 = \text{Synd}(\mathbf{x}_1)$,	8. Receives σ_0, σ_1 . Corrects $\hat{\mathbf{x}}_b = \text{Correct}(\hat{\mathbf{x}}_{ I_b}, \sigma_b)$.

***Reed–Solomon vs
low-density parity-check (LDPC) codes***

Quantum Oblivious Transfer (QOT)

Step 5: Encryption

Alice		Bob
9. Generates and sends seeds $\mathbf{s}_0, \mathbf{s}_1 \xleftarrow{\$} \{0, 1\}^{n_0 + \lambda_{PQS}}.$ Constructs and sends $E_0 = PRG(h_0(\mathbf{x}_0, \mathbf{s}_0)) \oplus m_0,$ $E_1 = PRG(h_1(\mathbf{x}_1, \mathbf{s}_1)) \oplus m_1.$	$\xrightarrow{\text{Classical}}$	9. Receives $\mathbf{s}_0, \mathbf{s}_1, E_0, E_1$ Decrypts $m_b = PRG(h_b(\hat{\mathbf{x}}_b, \mathbf{s}_b)) \oplus E_b.$

***Carter–Wegman vs
Toeplitz hash***

QOT Optimal Configuration – Error correction

- LDPC Error correction outperforming RS
- Clear turning point at 0.3-0.4 ->
 - From literature known that higher scores are possible

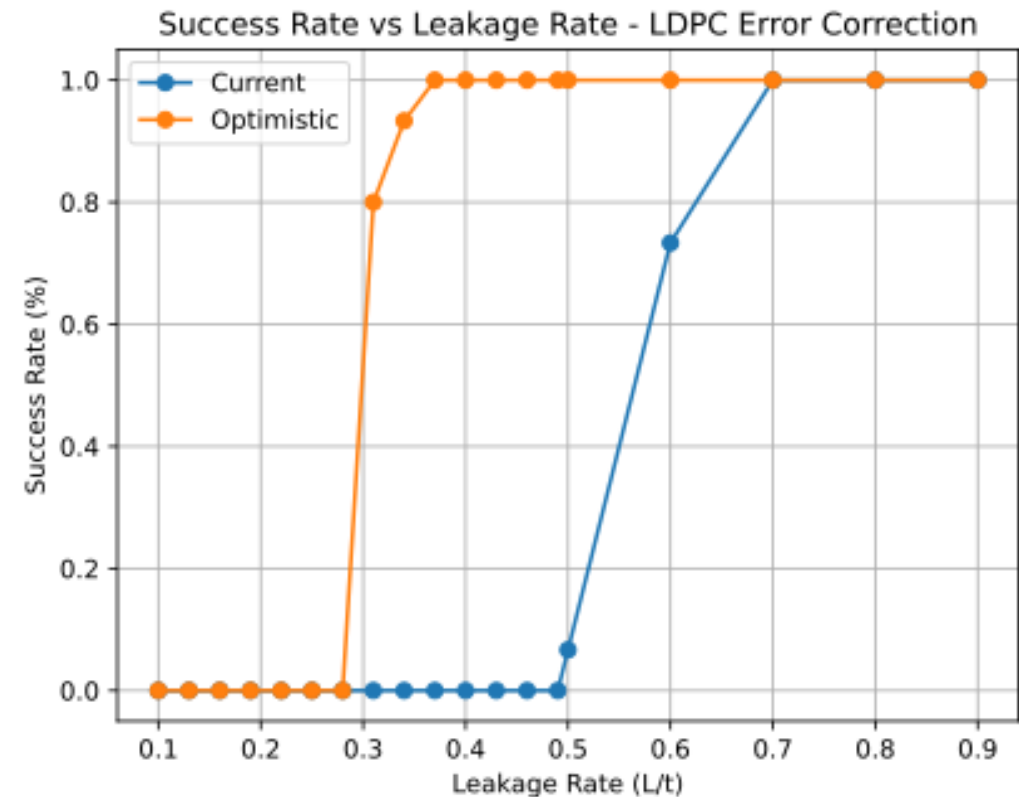
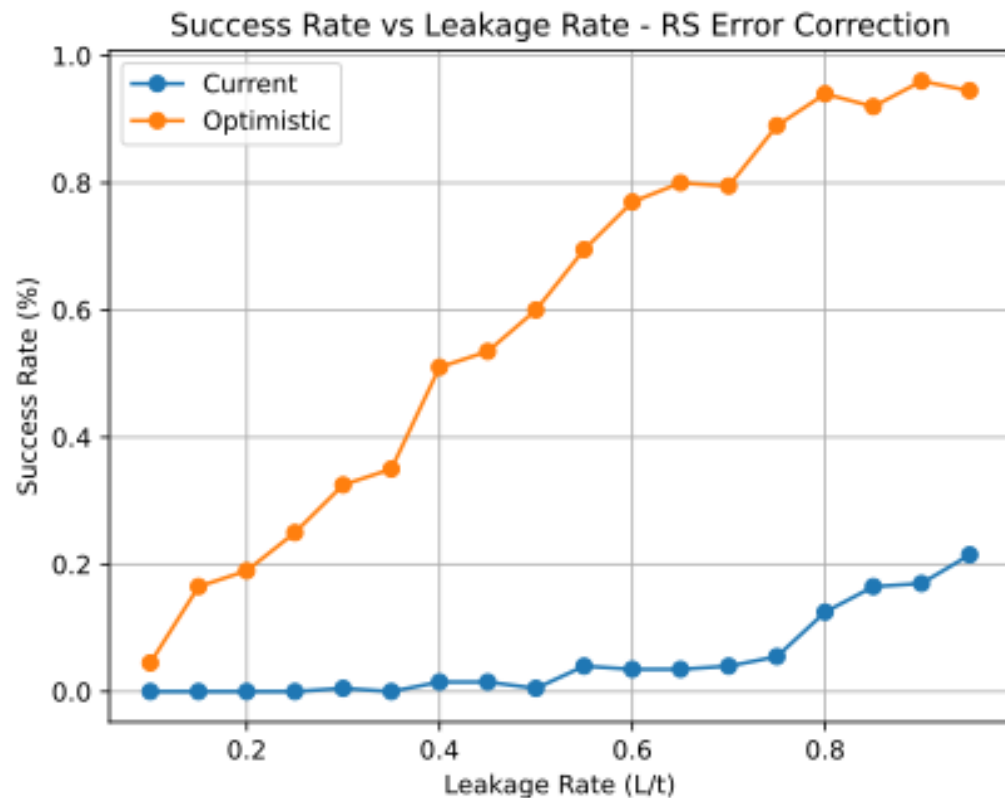


Figure 5.2: Comparison between success rates for the current and optimistic scenarios as a function of the leakage rate. For each data point the parameter. Showing RS (left) with $\lambda_{OT} = 256$, and LDPC error correction (right) with $\lambda_{OT} = 7680$

QOT – Performance

- Carter-Wegman hashing scales better than Toeplitz
- Post processing remains insignificant compared to QCommunication

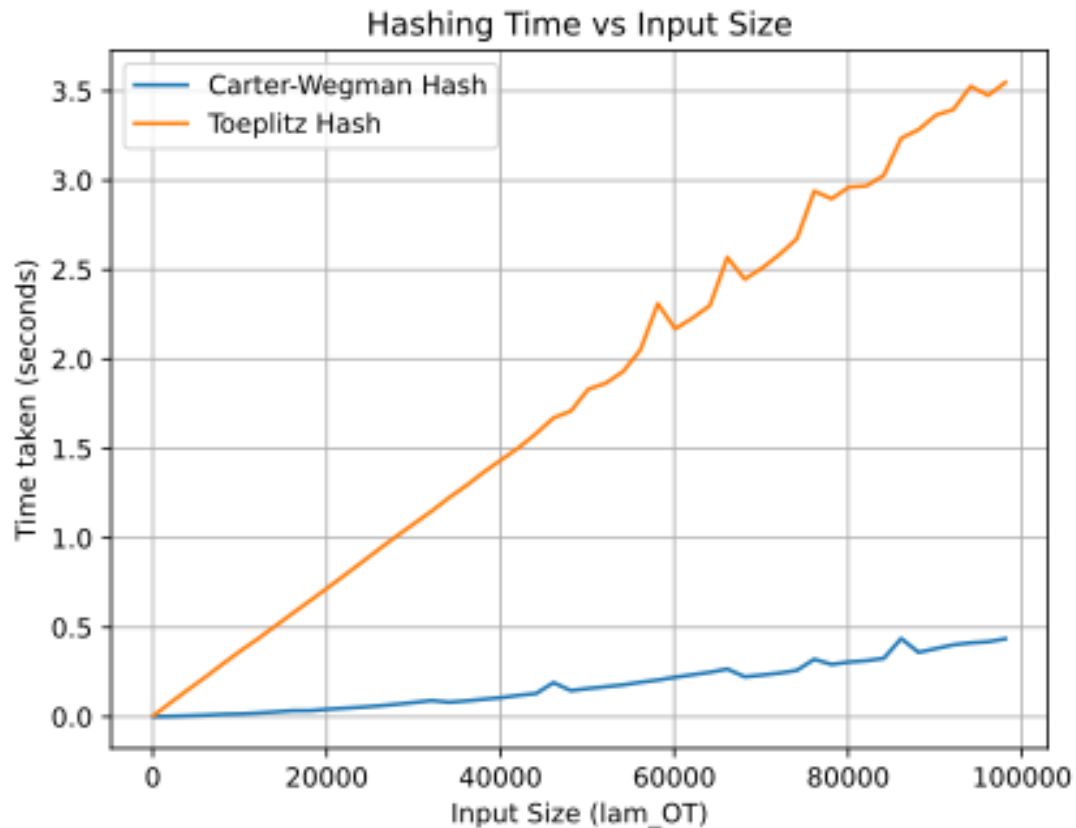


Figure 5.4: Measured time taken to hash a random bitstring with a given length for the two different hashes.

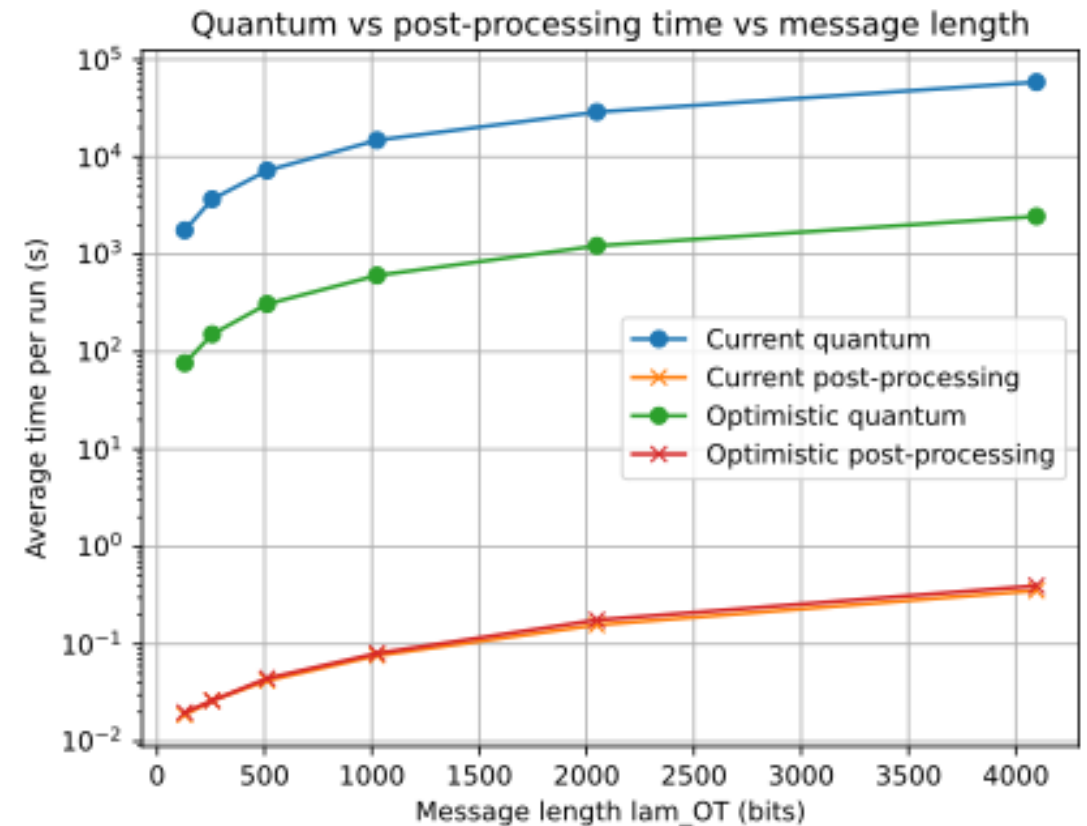


Figure 5.5: Time taken from the start of the protocol until the attempt at decoding the chosen message.

QOT – Performance

- Scales linearly for sufficient Leakage Rate,
- $\lambda_{OT} = 6000$ per hour, so 200 EPR pairs per minute
- Obliterated by Qline, which performs 1.4×10^4 times better.

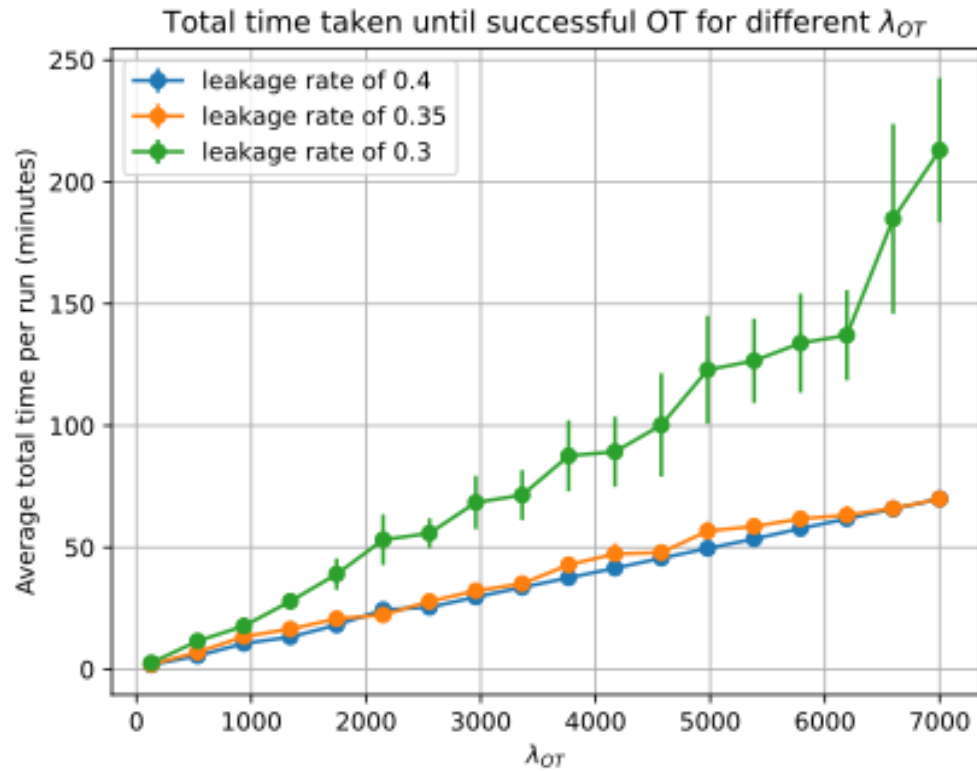


Figure 5.6: Our average time needed to obtain a successful 1-2 QOT execution.

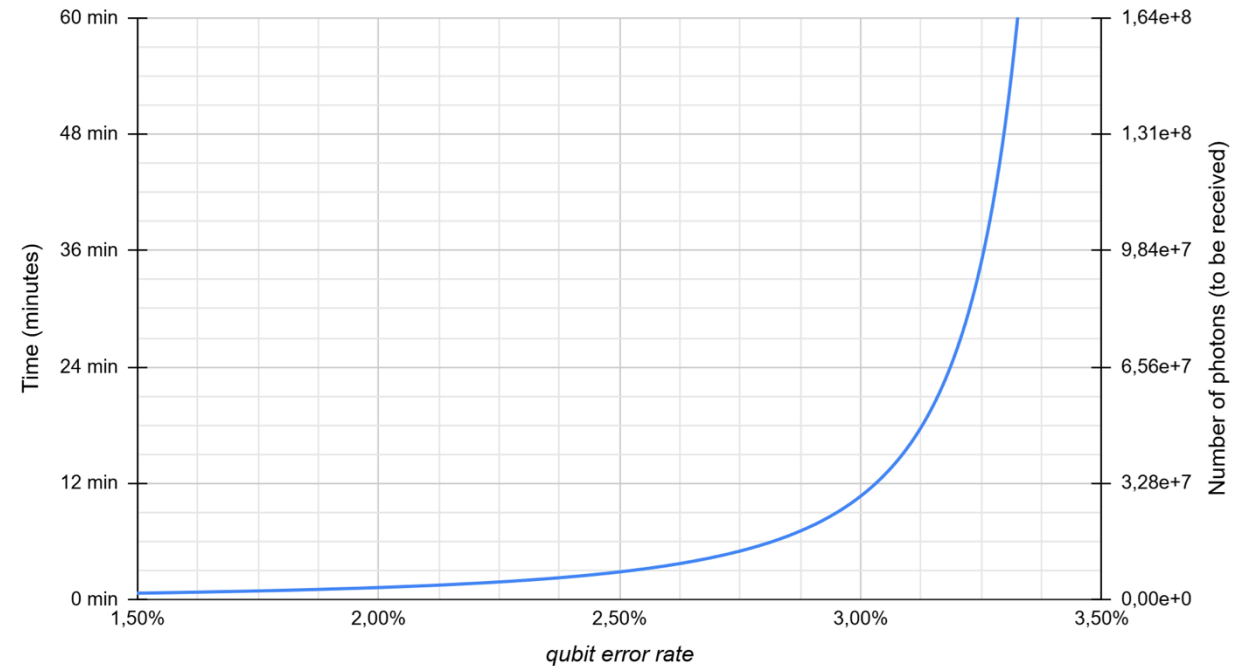


Figure 5.7: Estimation of the number of qubits and corresponding time to perform one OT on Qline

QOT – Error Bound

$$\Delta(\rho_{K_{\bar{b}}K_bE}, \frac{1}{2^l}I \otimes \rho_{K_b} \otimes \rho_E) \leq \frac{1}{2} \times 2^{-\frac{1}{2}} \left[\left(\frac{1}{2} - \xi - h(Q_{\text{tol}} + \delta) \right) \frac{\lambda_{OT}}{2} - q - l \right] + \sqrt{6} \exp\left(-\frac{\delta^2}{100} \lambda_{OT}\right) + 2 \exp\left(-\frac{\xi^2}{2} \lambda_{OT}\right).$$

- Difference between *real* and *ideal* state of security
- Calculate minimum λ_{OT} for a set security parameter ($\epsilon_{\text{sec}} = 2^{-22}$) and 3 leakage rates
- Minimum λ_{OT} of 4×10^6
- Comes down to weeks of runtime for one OT

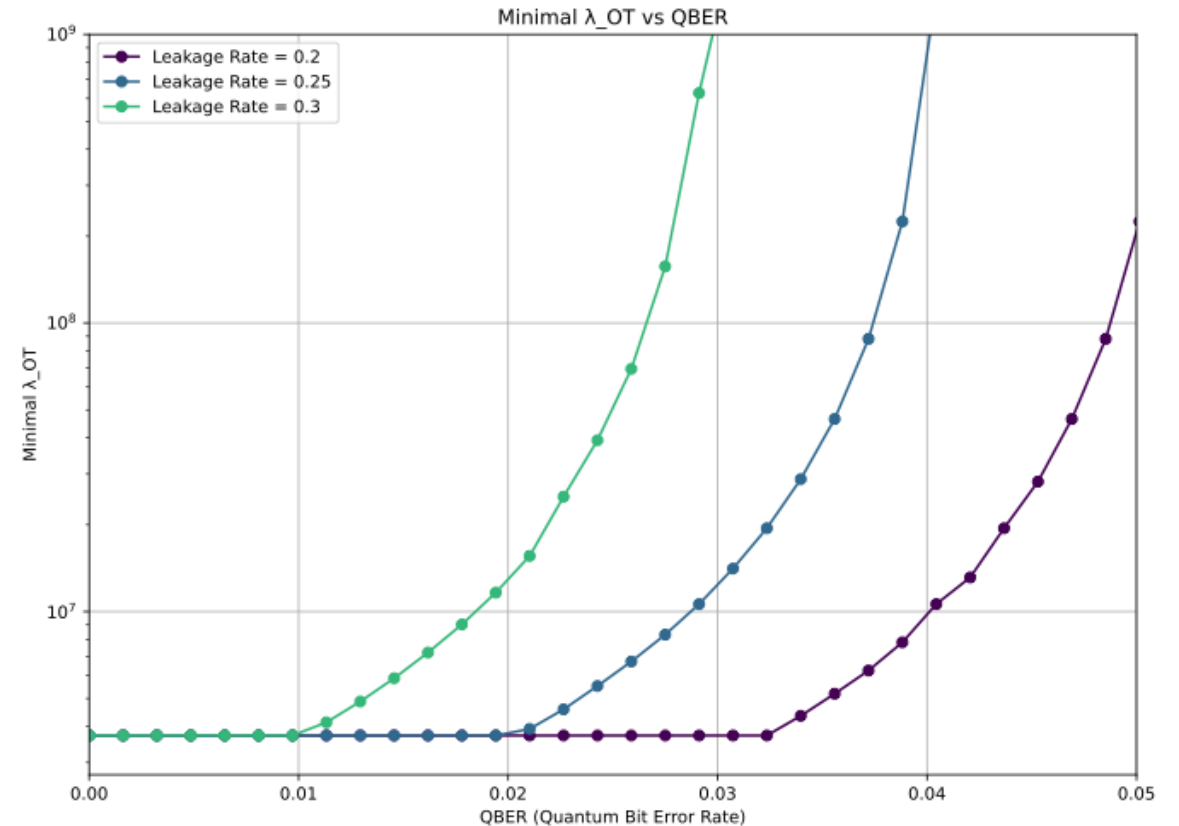


Figure 5.8: Minimal λ_{OT} as a function of the QBER for different leakage rates λ_{ϱ} .

QOT – Variants Performance

Bit/Base OT	1 QOT
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1-2 String OT	1 QOT
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1-N String OT	$\log_2(N)$ calls to QOT, potential for $\log(N)$
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Rabin OT	Set N based on erasure probability
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Main Findings

Successfully implemented 1-out-of-2 quantum oblivious transfer protocol in an entanglement-based quantum-network simulator

- LDPC codes outperform RS and Carter–Wegman outperforms Toeplitz hashing.
- Still, for optimistic scenario and $\epsilon_{\text{sec}} = 2^{-22}$, transfer would take over a month.
- Variants of OT for MPC don't add significant overhead

To conclude, getting rid of PKI does not justify the time requirements associated with QOT. Outperformed by cOT and Prepare and measure.